

Introduction

This application note provides guidance on how to configure the LAN8670/1/2 Ethernet PHY for optimal performance in 10BASE-T1S networks. Microchip is constantly developing improvements to the configuration. For the most current specifications, please visit the product pages on the website for LAN8670, LAN8671, and LAN8672 [10BASE-T1S Ethernet PHY Transceivers \[1\]](#).

The information presented in this application note is based upon the LAN8670/1/2 hardware revisions listed in the following table and is applicable for these and newer versions of the device until superseded by a newer version of the Configuration Application Note.

Table 1. LAN8670/1/2 Hardware Revisions

Product Revision ¹	Silicon Revision ²
C2	0101
D0	0110

Notes:

1. The product revision is noted in the package top marking.
2. The Silicon Revision is located in the PHY_ID1 register in the Manufacturer's Revision Number field.

1. Accessing Configuration Registers

The configuration settings for the LAN8670/1/2 consist of a series of configuration register accesses performed through the Serial Management Interface (SMI). The register accesses must be written immediately following a power cycle or reset of the device and must be performed in the sequence provided in this application note. Details for accessing the configuration registers through the SMI may be found in the device data sheet.

The types of register functions utilized in the configuration include:

- W – Write of value to specified register
- R – Read of value from specified register
- IR – Indirect read of value from specified register

Read Pseudocode

```
(uint16) value = read_register(uint8 mmd, uint16 addr)
```

Write Pseudocode

```
write_register(uint8 mmd, uint16 addr, uint16 value)
```

where:

mmd	MDIO Managed Device address of register
address	Address of the register to access
value	Value to be written to or read from the register at address

Indirect Read Pseudocode

Proprietary access mechanism. Do not confuse this with the Clause 22 indirect access to Clause 45 registers.

```
uint8 indirect_read(uint8 addr, uint8 mask)
{
    write_register(0x1F, 0x00D8, addr)
    write_register(0x1F, 0x00DA, 0x2)
    return (read_register(0x1F, 0x00D9) & mask)
}
```

2. Revision D0 and later - Configuration Process

The following configuration for the LAN8670/1 for silicon revision D0 and later, until superseded by another revision of this document, is to improve EMI performance, communication robustness and full compliance to OPEN Alliance specifications. It is advised to apply this configuration. Nonetheless, the device is still able to communicate rudimentarily when connected to an Ethernet network without this configuration.

2.1. Write Configuration into Registers (Optional)

Table 2-1 lists a series of parameters that must be configured with the given values and in the order listed. These settings must be configured for optimal performance and compliance to OPEN Alliance specifications.

Among the parameters is the Link Status Control (LSCTL, Address=0x0012) Register, which was introduced in silicon revision D0 and is defined in the *LAN8670/1/2 10BASE-T1S Ethernet PHY Transceiver Datasheet*. This register defines how link status is derived on the PHY. The user may decide which of the described methods best suits their application and which pin output may signal the current link status by writing the corresponding settings into the register.

Table 2-1. Configuration Register Writes

Access	MMD	Address	Writes
W	0x1F	0x0037	0x0800
W	0x1F	0x008A	0xBFC0
W	0x1F	0x0118	0x029C
W	0x1F	0x00D6	0x1001
W	0x1F	0x0082	0x001C
W	0x1F	0x00FD	0x0C0B ¹
W	0x1F	0x00FD	0x8C07 ¹
W	0x1F	0x0091	0x9660
W	0x1F	0x0012 (LSCTL)	0xFFFF ²

Notes:

- It is important to write both values shown in this table to this register, one after the other.
- The user may select how Link Status is derived and which pin indicates its status.
 - To provide backwards compatibility with earlier revision devices, the same behavior as earlier revision devices can be achieved in revision D0 devices and later by selecting no pin output and forcing Link Status to always return a '1' by writing '0x1001' into the LSCTL register.

2.2. PLCA Configuration Process - Revision D0 and later

Under normal operation, the device should be operated in PLCA mode.

Enabling PLCA

When using PLCA, the node with the Node_ID 0 is called the PLCA coordinator. For the PLCA coordinator, the number of transmit opportunities, which is the maximum Node_ID used in the mixing segment plus 1, must be written into the PLCA Control 1 (PLCA_CTRL1, Address = 0xCA02) register.

Nodes with IDs ranging between 1 and 0xFE are called PLCA followers. These nodes only require their Node_ID to be written into the field of PLCA_CTRL1.

Once the number of transmit opportunities and Node_ID have been written, enable PLCA by writing a '1' into the PLCA Enable (EN) bit in the PLCA Control 0 (PLCA_CTRL0, Address = 0xCA01) register.

The following code configures the device as either a PLCA coordinator or PLCA follower depending on the device's corresponding Node ID.

```
uint8 Node_ID
uint8 Node_Count
uint16 plcaparam1
Node_Count = MAX_NODE_ID + 1
if(Node_ID == 0)
{
    plcaparam1 = (uint16) (Node_Count << 8)
}
else
{
    plcaparam1 = (uint16) Node_ID
}
```

When using PLCA, it is also recommended to allow the PLCA status to determine whether collision detect is enabled or not. This can be done by writing a '1' into the Collision Detection Auto Disable (CDAD) bit of the Collision Detection Control 0 (CDCTL0, Address = 0x0087) register. In this mode collision detection is automatically disabled in PLCA Status is true and automatically enabled in CSMA/CD mode when PLCA Status is false. This setting provides the best performance in noise environments in PLCA mode but ensures proper operation in CSMA/CD mode.



When writing to CDCTL0, use a read-modify-write operation to avoid accidental modifications to reserved fields.

This sequence is summarized in [Table 2-2](#).

Table 2-2. PLCA Configuration Register Writes

Access	MMD	Address	Value
W	0x1F	0xCA02 (PLCA_CTRL1)	plcaparam1
W	0x1F	0xCA01 (PLCA_CTRL0)	0x8000
R	0x1F	0x0087 (CDCTL0)	CDCTL0_TEMP
W	0x1F	0x0087 (CDCTL0)	CDCTL0_TEMP 0x0200

Disabling PLCA

In some cases, it may be desired to force the device to remain in CSMA/CD mode. To do this, disable PLCA in the PLCA_CTRL0 register. Collision detect should be enabled by setting the Collision Detect Enable (CDEN) of the CDCTL register.

Table 2-3. Return to CSMA/CD from PLCA

Access	MMD	Address	Value
R	0x1F	0x0087 (CDCTL0)	CDCTL0_TEMP
W	0x1F	0x0087 (CDCTL0)	CDCTL0_TEMP (0x0200)
W	0x1F	0xCA01 (PLCA_CTRL0)	0x0000

3. Revision C2 - Configuration Process

Before configuring the LAN8670/1/2, some device configurations must be read and saved for proper configuration of the device.

3.1. Calculation of Configuration Parameters

Value1 has an expected range from -5 to 15. Value1 is stored as a signed 5-bit value in address location 0x0004.

```
uint8 value1 = indirect_read(0x04,0x1F)
int8 offset1
if((value1 & 0x10) != 0)
{
    offset1 = (int8) ((uint8)value1 - 0x20)
}
else
{
    offset1 = (int8) value1
}
```

Value2 is stored as a signed 5-bit value in address location 0x0008.

```
uint8 value2 = indirect_read(0x08,0x1F)
int8 offset2
if((value2 & 0x10) != 0)
{
    offset2 = (int8) ((uint8)value2 - 0x20)
}
else
{
    offset2 = (int8) value2
}
```

The following parameters must be calculated from the device configuration parameters mentioned above to use for the configuration of the registers.

```
uint16 cfgparam1 = (uint16) (((9 + offset1) & 0x3F) << 10) | (uint16) (((14 + offset1) & 0x3F) << 4) | 0x03
```

```
uint16 cfgparam2 = (uint16) (((40 + offset2) & 0x3F) << 10)
```

3.2. Write Configuration into Registers

[Table 3-1](#) lists a series of parameters that must be configured with the given values and in the order listed.

Table 3-1. Configuration Register Writes

Access	MMD	Address	Writes
W	0x1F	0x00D0	0x3F31
W	0x1F	0x00E0	0xC000
W	0x1F	0x0084	cfgparam1
W	0x1F	0x008A	cfgparam2
W	0x1F	0x00E9	0x9E50
W	0x1F	0x00F5	0x1CF8
W	0x1F	0x00F4	0xC020
W	0x1F	0x00F8	0xB900
W	0x1F	0x00F9	0x4E53
W	0x1F	0x0081	0x0080
W	0x1F	0x0091	0x9660

3.3. Write SQL Configuration into Registers (Optional)

If the SQL feature is to be used, the following configuration parameters must be calculated and saved so that they can be written into the corresponding registers in the next step.

```
uint16 cfgparam3 = (uint16) (((5 + offset1) & 0x3F) << 8) | (uint16) ((9 + offset1) & 0x3F)
```

```
uint16 cfgparam4 = (uint16) (((9 + offset1) & 0x3F) << 8) | (uint16) ((14 + offset1) & 0x3F)
```

```
uint16 cfgparam5 = (uint16) (((17 + offset1) & 0x3F) << 8) | (uint16) ((22 + offset1) & 0x3F)
```

Table 3-2 contains the registers that must be configured with the parameters listed and in the order given.

Table 3-2. SQL Configuration Register Writes

Access	MMD	Address	Value
W	0x1F	0x00AD	cfgparam3
W	0x1F	0x00AE	cfgparam4
W	0x1F	0x00AF	cfgparam5
W	0x1F	0x00B0	0x0103
W	0x1F	0x00B1	0x0910
W	0x1F	0x00B2	0x1D26
W	0x1F	0x00B3	0x002A
W	0x1F	0x00B4	0x0103
W	0x1F	0x00B5	0x070D
W	0x1F	0x00B6	0x1720
W	0x1F	0x00B7	0x0027
W	0x1F	0x00B8	0x0509
W	0x1F	0x00B9	0x0E13
W	0x1F	0x00BA	0x1C25
W	0x1F	0x00BB	0x002B

3.4. PLCA Configuration Process - Revision C2

Under normal operation, the device should be operated in PLCA mode. Disabling collision detection is recommended to allow the device to operate in noisy environments or when reflections and other inherent transmission line distortion cause poor signal quality. Disabling collision detection is also recommended during initial bring-up and debug. If the device determines that PLCA mode is not possible, it will revert into a fallback mode where PLCA is disabled and communication is handled in CSMA/CD mode. In this case, collision detection must be re-enabled.

Enabling PLCA

When using PLCA, the node with the Node_ID 0 is called the PLCA coordinator. For the PLCA coordinator, the number of transmit opportunities, which is the maximum Node_ID used in the mixing segment plus 1, must be written into the PLCA Control 1 (PLCA_CTRL1, Address = 0xCA02) register.

Nodes with IDs ranging between 1 and 0xFE are called PLCA followers. These nodes only require their Node_ID to be written into the field of PLCA_CTRL1.

Once the number of transmit opportunities and Node_ID have been written, enable PLCA by writing a '1' into the PLCA Enable (EN) bit in the PLCA Control 0 (PLCA_CTRL0, Address = 0xCA01) register.

The following code configures the device as either a PLCA coordinator or PLCA follower depending on the device's corresponding Node ID.

```
uint8 Node_ID
uint8 Node_Count
uint16 plcaparam1
Node_Count = MAX_NODE_ID + 1
if(Node_ID == 0)
{
    plcaparam1 = (uint16) (Node_Count << 8)
}
else
{
    plcaparam1 = (uint16) Node_ID
}
```

In many cases, as described above, collision detection should be disabled by writing a '0' into the Collision Detection Enable (CDEN) bit of the Collision Detection Control 0 (CDCTL0, Address = 0x0087) register.



When writing to CDCTL0, use a read-modify-write operation to avoid accidental modification of reserved fields.

This sequence is summarized in [Table 3-3](#).

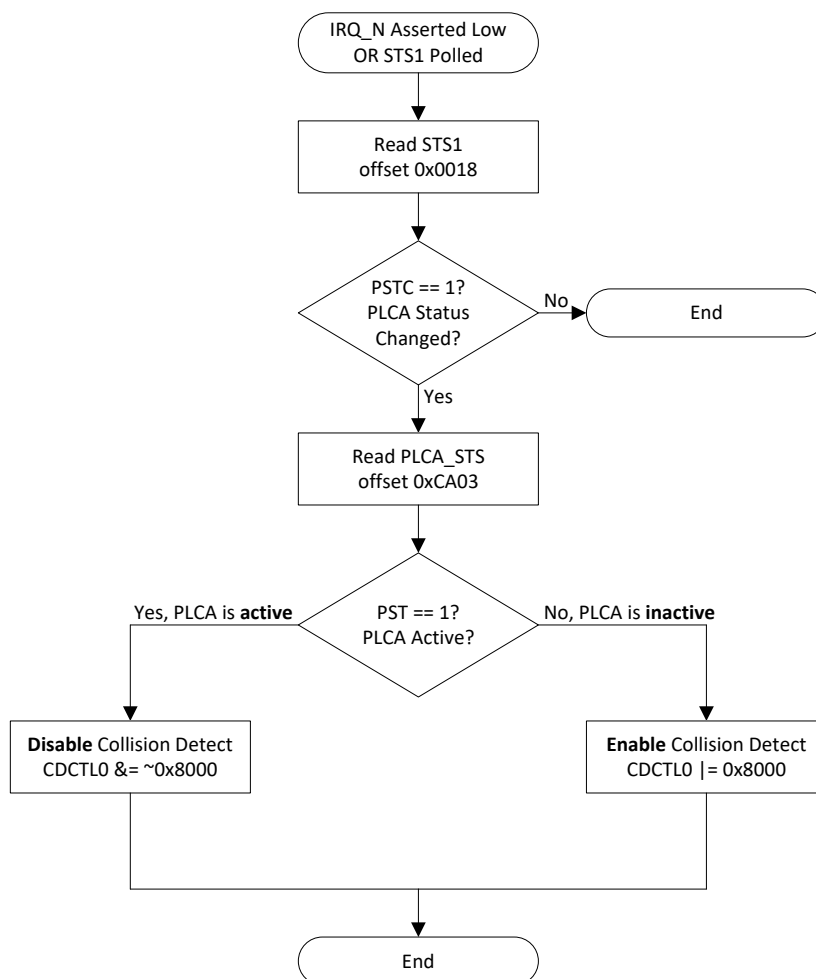
Table 3-3. PLCA Configuration Register Writes

Access	MMD	Address	Value
W	0x1F	0xCA02 (PLCA_CTRL1)	plcaparam1
W	0x1F	0xCA01 (PLCA_CTRL0)	0x8000
R	0x1F	0x0087 (CDCTL0)	CDCTL0_TEMP
W	0x1F	0x0087 (CDCTL0)	CDCTL0_TEMP & (~0x8000)

Managing Collision Detection

When configured for PLCA operation, it may be advantageous to disable collision detection. Should the device not detect the periodic presence of a PLCA BEACON, it will automatically fall back to CSMA/CD operation to keep communications running. Whenever PLCA is active, the PLCA Status (PST) bit in the PLCA Status (PLCA_STS, Address = 0xCA03) register is set by hardware and changes from a '0' to a '1'. Similarly, whenever the PLCA BEACON is lost and the device falls back to CSMA/CD operation, the bit changes from a '1' to a '0'. Whenever these changes occur, a '1' is set in the PLCA Status Changed (PSTC) bit in the Status 1 (STS1, Address = 0x0018) register. This flag is used to indicate that the activity status of PLCA has changed since the last read of the register.

Figure 3-1. PLCA Collision Detect Flowchart



When disabling collision detection while PLCA is active, it is necessary to manually re-enable it during the CSMA/CD fallback. The best way to do this is to enable an interrupt to notify the microcontroller whenever a PLCA status change has occurred. This is done by writing a '0' into the PLCA Status Changed Mask (PSTCM) bit in the Interrupt Mask 1 (IMSK1, Address = 0x001C) register. Alternatively, if no interrupts are being used, the device can be polled regularly. This can, however, lead to reduced performance.

Whenever an interrupt is triggered through the IRQ_N pin or a poll occurs, the STS1 register should be read. If the PLCA Status Changed (PSTC) bit is set, then collision detection should be enabled or disabled as shown in [Figure 3-1](#). During the handling of this interrupt or poll, the PLCA_STS register will indicate if PLCA is currently active or not. If PLCA is currently active, as indicated by reading the PLCA Status bit as '1', collision detection is disabled by clearing the Collision Detection Enable (CDEN) bit in the Collision Detect Control 0 (CDCTL0, Address = 0x0087) register. When the PLCA Status bit is read as '0', indicating that PLCA is inactive and the device is operating in CSMA/CD, then collision detection is enabled by writing a '1' to the CDEN bit.

Disabling PLCA

In some cases, it may be desired to force the device to remain in CSMA/CD mode. To do this, disable PLCA in the PLCA_CTRL0 register and enable collision detect in the CDCTL0 register.

Table 3-4. Return to CSMA/CD from PLCA

Access	MMD	Address	Value
R	0x1F	0x0087 (CDCTL0)	CDCTL0_TEMP
W	0x1F	0x0087 (CDCTL0)	CDCTL0_TEMP (0x8000)
W	0x1F	0xCA01 (PLCA_CTRL0)	0x0000

4. References

This application note is intended to be used in conjunction with the following products and documents:

1. LAN8670, LAN8671, and LAN8672 10BASE-T1S Ethernet PHY Transceivers
www.microchip.com/en-us/product/LAN8670
www.microchip.com/en-us/product/LAN8671
www.microchip.com/en-us/product/LAN8672
2. LAN8670/1/2 10BASE-T1S Ethernet PHY Transceiver Data Sheet, DS60001573
www.microchip.com/LAN8670

Note: *: A link to this application note can be found on the product pages of all applicable components.

5. Revision History

Table 5-1. Document Revision History

Revision	Changes
Current Document Revision	
Revision G (DS60001699G - October 2025)	Added configuration for silicon revision D0 Clarified which configurations are required for which silicon revision
Previous Document Revision	
Revision F (DS60001699F - June 2024)	Updated register configuration process Clarified calculation of configuration parameters
Revision E (DS60001699E - December 2023)	Updated register configuration process Clarified calculation of configuration parameters
Revision D (DS60001699D - November 2023)	Updated configuration register writes to clarify SQL configuration Updated PLCA and CSMA/CD configuration
Revision C (DS60001699C - July 2023)	Updated to new silicon revision
Revision B (DS60001699B - July 2023)	Updated to new silicon revision
Revision A (DS60001699A - July 2021)	Initial Release

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